

```

breaklines=true,
breakatwhitespace=false,
columns=fullflexible,
keepspaces=true,
frame=single,
framesep=2pt,
showstringspaces=false,
backgroundcolor=codebg,
rulecolor=codeframe,
keywordstyle=keyword,
stringstyle=string,
commentstyle=comment,
numberstyle=gray,
numbers=left,
numbersep=6pt,
xleftmargin=14pt,
framexleftmargin=14pt,
literate=

```

```

aa1
ee1
zz1
ss1
ff1
cc1
nn1
oo1
zz1

```

```

→1
←1
≠1

```

```

qkd/proofs/tight.py
3.7.6-fixed
εsec
TallyCounts
DecoyEstimatesInput
yield_1_lower_bound
error_rate_1_upper_bound
KeyCalculationResult
Lim2014Proof
qkd/proofs/lim2014.py
allocate_ε
calculate_key_length
Lim2014Proof
BB84TightProof
Lim2014Proof

```

```

X
Y1
e1
X
Z
nX
mX
nZ
mZ
sX,1L
sZ,1L
εsec
εsec
εsec
εsec
εsec

```

$$(1) \quad \varepsilon_{sec} \geq \varepsilon_{PE} + 2\varepsilon_{smooth} + \varepsilon_{PA} + \varepsilon_{cor} + \varepsilon_{phaseest}.$$

```

εPE
εsmooth
εPA
εcor
εphaseest
tight.py
allocate_ε
allocation.validate()
εcor +
(εsec -
εcor)
εsec
10-20
ZX

```

$$e_{phase} \leq e_{bit} + \sqrt{\frac{-\ln(\varepsilon_{phaseest})}{2s_{Z,1}^L}},$$

$$(2) \quad e_{bit} = \frac{\varepsilon_{phaseest}}{s_{Z,1}^L} \\ Z \\ \text{_compute_phase_error_ub} \\ \text{_assume_phase_equals_bit_error} \\ e_{phase} = e_{bit} \\ Z \\ H_2(p) \\ H_2(p) = -p \log_2(p) - (1-p) \log_2(1-p).$$

$$(3) \quad p =$$